Zach Scroggins

Senior Full Stack TypeScript Engineer

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PROFESSIONAL SUMMARY

Senior Full stack TypeScript engineer specializing in React, Next.js and Node.js, with a strong AWS background and hands-on experience building AI applications. Rose to Senior Software Engineer in a single year at Lumeris, a healthcare technology company. There I led the front-end framework evaluation that set the standard for new web applications, co-led development of an AI application that surfaced patient information alongside physicians' Electronic Medical Record (EMR), and helped migrate the primary analytics database to AWS. Brings an eye for UI design and user experience, and explains complex technical concepts clearly to non-engineers.

EXPERIENCE

Software Engineering Consultant | June 2025 – Present

TypeScript, Next.js, React Native

- Designed and built web and mobile applications with Next.js and React Native, including a fly fishing trip tracker, a game server hosting platform and a mountain real estate site.

Lumeris (Healthcare Technology) | January 2022 – January 2025

TypeScript, AWS, Infrastructure as Code, NestJS, SvelteKit, Angular, Vue, Electron, Azure OpenAI, Atlassian and Azure CI/CD, Scrum, Test-Driven Development

Senior Software Engineer (2 years)

- Co-led development of an AI value-based care application that worked alongside physicians' existing EMR to identify patients and surface information that might otherwise be missed. Helped prevent two potential ER visits during beta testing.
  - Built with OCR and LLMs via AWS Textract and Bedrock, embeddings in a Pinecone vector database, confidence score calibration and prompt engineering.
- Lead developer of an internal generative AI application on Azure OpenAI, giving employees access to generative AI while meeting the company's healthcare data privacy requirements.
- Led the front-end framework evaluation: researched the options, built comparable prototypes, assessed each against the company's needs and presented findings to

technology leadership. The recommendation set the company's move from Angular to SvelteKit.
- Took on technical leadership after promotion, supervising work of other engineers and taking part in higher-level technical decisions.
- Proposed and built a shared component library after seeing teams duplicate work across the existing Vue application and the newer Electron and SvelteKit application; now shared across all three.
- Created an infrastructure-as-code template for deploying SvelteKit on AWS Lambda and S3.
- Modernized the customer support database dashboard and built an internal AWS RDS management dashboard.

Software Engineer (1 year)

- Migrated Lumeris' primary analytics database from on-premises infrastructure to AWS RDS, working as a two-person team with an application architect on a business-critical system.
- Led the AWS migration of the public document service application (10,000+ documents), setting the example for using AWS Lambda@Edge with the CloudFront CDN.
- Prototyped a real-time analytics engine on AWS Kinesis, Lambda and HealthLake, a HIPAA-eligible service that stores, transforms and analyzes healthcare data using the FHIR R4 standard.

Freelance Full Stack Developer | August 2020 – December 2021

- **PaulitaCarlotta.com** (paid - not maintained by owner): e-commerce store for women's fashion and jewelry with a lifestyle blog. *JavaScript, Next.js, React.js, Node.js, BigCommerce API, Prismic CMS API.*
- **untcsce.club** (pro bono, UNT Computer Science Club): informational and marketing site with a headless CMS. *TypeScript, Next.js, React.js, Node.js, Commerce.js, Stripe, Contentful CMS API.*

RECENT TECHNICAL PROJECTS

NASA API Explorer

TypeScript, Next.js, React.js, Node.js, REST API, NASA API, Claude Code

- Modernizing the codebase with Claude Code: directing the AI's work, then manual and AI peer review.
- Searches NASA's official database of over 140,000 photos of the universe.
- Shows satellite imagery from the Earth Polychromatic Imaging Camera, with positional data calculated using J2000 coordinates, vector calculus and trigonometry.

zachscroggins.com

TypeScript, Next.js, React.js, Node.js, GitHub API

- Designed and built a portfolio site that showcases my professional experience and projects, recently updated with my Lumeris work.
- Demonstrates both front-end development in React and Next.js and my own UI design.

SKILLS

Languages: TypeScript, JavaScript, HTML, CSS, Python, Java, C#

Front End: React, Next.js, SvelteKit, Angular, Vue, Electron, Tailwind CSS

Back End & Cloud: Node.js, NestJS, AWS (Lambda, S3, RDS, CloudFront, Kinesis, Bedrock, Textract), Infrastructure as Code, PostgreSQL, Microsoft SQL, REST and GraphQL APIs, Docker

AI: AWS Bedrock, Azure OpenAI, OpenAI API, Claude Code, GitHub Copilot, vector databases (Pinecone), prompt engineering

Testing & Tools: Playwright, Cypress, Vitest, Jest, Git, GitHub, CI/CD, Vercel

Design: UI and UX design, Figma, Adobe Suite

EDUCATION

University of North Texas, Denton, TX | January 2021 – December 2021

Computer Science Major

- Web Development Officer, Computer Science Club
- Coursework: CS II, Data Structures, Algorithms, Usability and UX

North Central Texas College, Corinth, TX | June 2019 – December 2020

Computer Information Systems Major

- Coursework: CS I, OOP, Java, C#, LAMP Stack, Linux, Web Design, PC Hardware, Computer Networks, Computer Graphics, Digital Imaging